

ALGEBRA

Basic Math

Quadratic Equation

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- Discriminant

$$B^2 - 4AC$$

o = 0; equal

o > 0; unequal

o < 0; imaginary

- Properties

o sum

$$x_1 + x_2 = \frac{-B}{A}$$

o product

$$x_1 x_2 = \frac{C}{A}$$

Binomial Theorem

- r^{th} term in $(a+b)^n$
 $r^{\text{th}} = nCr a^{n-m} b^m$
 o $m = r - 1$

$$nCr = \frac{n!}{(n-m)!m!}$$

- constant term

o equate exponent of a to 0

o sum of the coefficients

$$S = f(1) - f(0)$$

o sum of exponents

$$Se = \frac{n}{2}(n+1)(j+k)$$

Other Formulas

- Laws of Exponents

$$1. (a^m)^n = a^{mn}$$

$$2. a^m \times a^n = a^{m+n}$$

$$3. \frac{a^m}{a^n} = a^{m-n}$$

$$4. a^{-n} = \frac{1}{a^n}$$

$$5. (a \times b)^m = a^m \times b^m$$

$$6. \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

- Properties of Logarithm

$$1. \log_a MN = \log_a M + \log_a N$$

$$2. \log_a \frac{M}{N} = \log_a M - \log_a N$$

$$3. \log_a M^n = n \log_a M$$

$$4. \log_a a = 1$$

$$5. \log_a a^x = x \log_a a = x$$

$$6. \log_a 1 = 0$$

$$7. \text{If } \log_a M = N, \text{ then } a^N = M$$

$$8. \text{If } \log_a M = \log_a N, \text{ then } M = N$$

$$9. \log_e M = \ln M$$

$e = 2.71828...$ (Napierian logarithm)

$$10. \log_{10} M = \log M \text{ (Common logarithm)}$$

$$11. \log_n M = \log M / \log n = \ln M / \ln n$$

$$12. \text{If } \log_b x = a \text{ then } x = \text{antilog}_b a$$

$$13. a^x = \text{antilog}_a x$$

$$14. \log_{10} 4250 = \log_{10} (1000 \times 4.25)$$

$$= \log 1000 + \log 4.25$$

$$\log_{10} 4250 = 3 + 0.6284 = 3.6284$$

3, the integral part, is called the characteristic

0.6284, a non-negative decimal fraction part, is called the mantissa

Partial Fractions

- Case 1

$$\frac{3x + 2}{(x + 1)(x + 4)} = \frac{A}{x + 1} + \frac{B}{x + 4}$$

- Case 2

$$\frac{3x + 2}{(x + 1)(x^2 + 4)} = \frac{A}{x + 1} + \frac{Bx + c}{x^2 + 4}$$

- Case 3

$$\frac{3x + 2}{(x + 1)(x + 4)^2} = \frac{A}{x + 1} + \frac{B}{x + 4} + \frac{C}{(x + 4)^2}$$

Word Problems I

Sample of words used to represent mathematical operations:

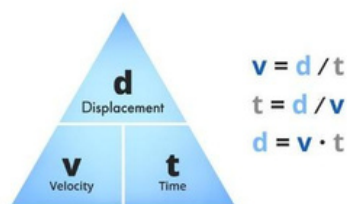
Add	Subtract	Multiply	Divide
add	subtract	multiply	divide
sum	difference	times	quotient
increased by	decreased by	product	per
more than	fewer	double	divided equally
total	less than	twice	split into
in all	diminished by		fraction
gain	reduced by		ratio of
plus	minus		

Age Problems

- **Tip 1:** If the current age is x , then n times the age is nx .
- **Tip 2:** If the current age is x , then the age n years later $= x + n$
- **Tip 3:** If the current age is x , then age n years ago $= x - n$
- **Tip 4:** The ages in a ratio $a:b$ will be ax and bx
- **Tip 5:** If the current age is x , then $1/n$ of the age is x/n
 - x = present
 - $x + n$ = n years later
 - $x - n$ = n years ago

Motion Problems

VELOCITY FORMULA



- motion in a water / air current
 - o x = velocity of boat / plane in still water / air
 - o y = velocity of water / air
 - o $x + y = v$ when going downstream / with the wind
 - o $x - y = v$ when going upstream / against the wind
- motion in a circle or any closed circuit

$$d_f - d_s = 1 \text{ circuit} \rightarrow \text{same direction}$$

$$d_f - d_s = 1 \text{ circuit} \rightarrow \text{opposite direction}$$

$$d_f = d_s \rightarrow \text{overtaking of faster}$$

Note:

- ♦ The age difference between two persons is constant.

Work Problems

- Case 1: Each worker has different rates

$$\text{rate} = \frac{\text{job done}}{\text{time}}$$

$$\text{combined rates} = \text{rate 1} \pm \text{rate 2}$$

- Case 2: Each worker has the same rate

$$\text{job done} = (\text{number of men})(\text{number of days})$$

- o to finish the job

$$WT = W_1T_1 + W_2T_2 + W_nT_n$$

$$\Sigma \text{job done} = \text{total number of man}$$

– days

$$\text{rate} = \frac{\text{job done}}{\text{time}}$$

combined rates

$$= \text{rate 1} \pm \text{rate 2}$$

$$\bullet \frac{1}{T} = \frac{1}{t_1} + \frac{1}{t_2}$$

- o t_1 = time taken by the

first person to finish the work

- o t_2 = time taken by the

second person to finish the work

- o T = total time taken if both persons work together

$$\bullet 1 = \frac{1}{A}x + \frac{1}{B}y$$

- o A days; x hours/days

- o B days; y hours/days

Clock Problems



- sx = distance of minute-hand
- $\frac{x}{12}$ = distance of hour-hand
- $1 \text{ hr} = 360^\circ$
- $12 \text{ hr} = 360^\circ$
- $60 \text{ min} = 360^\circ$
- $1 \text{ min} = 6^\circ$

Digit Problems

$$100h + 10t + u$$

- reversed

$$100u + 10t + h$$

Series and Sequences

Arithmetic Series

$$an = a_1 + (n-1)d$$

- sum

$$Sn = \frac{n}{2} (a_1 + a_n)$$

Geometric Series

$$an = a_1 r^{n-1}$$

- sum

o $r > 1$ (finite)

$$Sn = \frac{a_1(1-r)^n}{1-r}$$

o $r < 1$ (infinite)

$$Sn = \frac{a_1}{1-r}$$

Harmonic Progression

- reciprocal of arithmetic
- n^{th} term

$$a_n = \frac{1}{a_1 + (n-1)d}$$

- sum

$$Sn = \frac{1}{d} \ln \left(\frac{2a_1 + (2n-1)d}{2a_1 - d} \right)$$

Means

$$AM = \frac{x + y + \dots}{n} = \sum_{i=1}^n x_i$$

$$GM = \sqrt[n]{xy} = \sqrt[n]{(AM)(HM)} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

$$HM = \left(\frac{x^{-1} + y^{-1} + \dots}{n} \right)^{-1} = \frac{1}{AM}$$

Word Problems II

Remainder Theorem

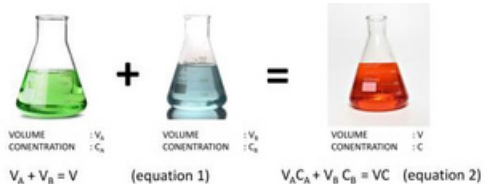
- remainder = $f(x)$

$r=0 \rightarrow$ factor

- Special Products and Factoring

1. $(x+y)(x-y) = x^2 - y^2$
2. $(x+y)^2 = x^2 + 2xy + y^2$
3. $(x-y)^2 = x^2 - 2xy + y^2$
4. $(x+y+z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$
5. $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$
6. $x^3 - y^3 = (x-y)(x^2 + xy + y^2)$
7. $x^6 - y^6 = (x^2)^3 - (y^2)^3 = (x^2 - y^2)[(x^2)^2 + (x^2)(y^2) + (y^2)^2]$
 $= (x+y)(x-y)(x^4 + x^2y^2 + y^4)$

Mixture Problems



Fundamental Principles of Counting

Fundamental Principle of Counting

$$m \times n$$

Permutation

- Theorem 1

o n different objects taken r at a time

$$P(n,r)=nPr=\frac{n!}{(n-r)!}$$

• Theorem 2

o n objects taken all at a time in which n_1 of the objects are alike, n_2 are alike, n_3 are alike, and so on

$$\frac{n!}{n_1!n_2!n_3!}$$

• Theorem 3

o n objects into r cells with n_1 elements in the first cell, n_2 elements in the second cell, and soon

$$\frac{n!}{n_1!n_2!n_3!}$$

Cyclic Permutation

- n objects in a circle taken r at a time

$$(nC_r)(r-1)! = \frac{nPr}{r}$$

- n objects in a circle taken r at a time

$$(n-1)!$$

Combination

• Theorem 4

o n different objects taken r at a time

$$C(n,r)=nC_r=\frac{n!}{r!(n-r)!}$$

• Theorem 5

o n objects taken taken 1 at a time, 2 at a time, 3 at a time, and so on until n at a time

$$2^n - 1 = \sum_{x=1}^n nCx$$

Note:

$$\diamond nPn = n!$$

$$\diamond nP0 = 1 \diamond$$

$$nCn = 1 \diamond$$

$$nC0 = 1 \diamond$$

$$0! = 1$$

Probability

$$P = \frac{\text{number of favorable outcomes}}{\text{number of possible outcomes}}$$

$$P + Q = 1$$

- P = probability to happen
- Q = probability to fail

Types of Events

- mutually exclusive events (+)
 - o impossible for more than one of them to happen in a single trial
- independent events (x)
 - o happening of one does not affect the probability of happening of the other
- dependent events (x)
 - o happening of one affects the probability of happening of the other

Expectation

$$E = \sum (\text{profit} \times \text{probability})$$